Basic Dense Matrix Multiplication in Python, Java and C: A Controlled Benchmarking Study

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Abstract

Question: how does matrix size n affect the runtime and memory of one straightforward $O(n^3)$ triple-loop dense matrix multiplication implemented in Python, Java and C under identical, controlled conditions? We implement the same i–j–k algorithm in the three languages with IEEE-754 `float64` arithmetic and byte-identical inputs (seed 42, $n \in \{10, 50, 100, 200, 500, 1000, 2000\}$), and measure the isolated multiplication kernel with monotonic clocks, warm-ups, batches of at least 0.2 s, five repetitions per configuration, and a declared budget of 60 s per multiplication and 4 GiB of memory. Across the feasible range, runtime grows close to the cubic model (fitted exponents 3.22–3.64 for $n \geq 100$) with large language-dependent constants: Python takes 113–169× longer than C and Java 1.2–2.9× longer than C. Peak resident memory grows quadratically plus a language runtime overhead. The largest feasible sizes under the budget are $n = 500$ for Python (24.2 s) and $n = 1000$ for Java (5.2 s) and C (2.3 s); $n = 2000$ exceeded the budget in all three languages and is reported as a timeout without an invented runtime. Conclusions are scoped to these implementations, this machine and these inputs.

1 Introduction and objective

Dense matrix multiplication is the canonical example for reasoning about algorithmic complexity: the textbook triple loop performs $2n^3$ floating-point operations, so any implementation should show roughly cubic growth with the side length n [1, 2]. In practice the constant factors differ by orders of magnitude between an interpreted language, a just-in-time compiled virtual machine and ahead-of-time compiled native code, and the memory hierarchy eventually dominates the arithmetic.

Research question. How does the matrix size n affect the runtime and the memory use of a basic dense matrix multiplication implemented in Python, Java and C, under the documented execution conditions of this study?

Hypotheses (stated before measurement).

H1: Runtime grows as $T(n) \propto n^3$ over the tested range, with measurable deviations at large n caused by the memory hierarchy.

H2: The constant factors are ordered Python $\gg$ Java $\approx$ C, because Python executes each scalar operation through the interpreter, HotSpot compiles Java bytecode to machine code, and C is compiled ahead of time.

H3: Memory grows as $3 \times 8n^2$ bytes for the three `float64` matrices plus a per-language runtime overhead.

The factor comparison is the controlled variable: same algorithm, same arithmetic, same input files, same timing boundary, same machine. The outcome measures are the kernel time per multiplication and the peak resident memory of the process.

2 Background

Algorithm. The straightforward (naive) multiplication of two $n \times n$ matrices $C = AB$ in row-major order is

```
for i in 0..n-1:
  for j in 0..n-1:
    s = 0
    for k in 0..n-1:
      s += A[i][k] * B[k][j]
    C[i][j] = s
```

which performs n^3 multiply–accumulate steps, i.e. $2n^3$ FLOPs. The loop order determines the summation order; fixing it across languages makes the floating-point results bitwise identical [7, 6].

Runtime systems. CPython executes each bytecode with a dispatch loop and boxes every matrix element as a `float` object, so element access and arithmetic are interpreted [6]. HotSpot compiles hot Java methods with a tiered JIT (interpreter → C1 → C2); measurements must exclude the compilation phase with warm-ups [5]. GCC with `-O2` produces ahead-of-time native code, including auto-vectorisation on x86-64 with the compiler's default instruction set; this is a fixed property of the documented flags, not a hand-written optimisation.

Memory model. A dense `float64` matrix occupies $8n^2$ bytes, so the three working matrices need $24n^2$ bytes: 0.24 MB at $n = 100$, 6 MB at $n = 500$, 24 MB at $n = 1000$ and 96 MB at $n = 2000$, to which interpreters, runtimes and allocator bookkeeping add a roughly constant overhead [1].

Benchmarking practice. We follow established guidance: use the median rather than the mean for skewed timing samples, report variability, keep raw measurements, control inputs, and never mix timing boundaries in one comparison [3, 4].

3 Methodology

3.1 Implementations

One triple-loop kernel per language, identical i–j–k order, no library calls inside the timed region, no blocking, no threading:

- **Python 3.11.9** – lists of row lists of Python `float` objects (`src/python/matmul.py`); pure interpreter, no NumPy in the kernel (NumPy is used only to generate inputs and as a correctness reference).
- **Java 21 (OpenJDK)** – `double[][]` row-major (`src/java/MatMul.java`); warmed with 20 000 calls at $n < 50$, 20 calls at $50 \le n \le 100$ and 5 calls at $n \ge 200$, all excluded from the statistics. The measurement runs add `-XX:CompileCommand=dontinline,MatMul::matmulInto` so that the JIT cannot eliminate the timed call as loop-invariant work inside the batch loop; the emitted checksum verifies that every result is consumed.
- **C11 (GCC 16.1.0, `-O2 -std=c11 -Wall -Wextra`)** – three matrices in one contiguous `double` allocation, row-major (`src/c/matmul.c`).

All three use IEEE-754 binary64 (`float64`) arithmetic in the same summation order, single-threaded (no numerical library threads are active inside the kernel).

3.2 Inputs and correctness

For every $n \in \{10, 50, 100, 200, 500, 1000, 2000\}$ a single pair of matrices A, B is generated with NumPy PCG64, seed scheme `default_rng([42, n, index])`, uniform values in $[-1, 1)$, stored as raw little-endian `<f8` files in row-major order, and shared byte-for-byte by the three languages; SHA-256 checksums are recorded in `data/inputs/meta.json`.

Before benchmarking, nine cases per language (hand-computed 3×3 product, identity-left/right, zero-left/right, two random sizes, two shared sizes) were compared against NumPy with $rtol = 10^{-9}$, $atol = 10^{-12}$, plus pairwise comparisons of the three languages: 55/55 comparisons passed, and all 27 pairwise comparisons were bitwise equal. Against NumPy, 15 of 27 results were bitwise equal and the remaining 12 differed only within tolerance because NumPy's SIMD summation order differs from the scalar triple loop – an expected evaluation-order effect.

3.3 Environment

Table 1: Execution environment (from `data/results/environment.json`).

Item	Value
CPU	13th Gen Intel Core i7-1355U (10 physical / 12 logical processors)
RAM	16.8 GB (16 GiB)
OS	Windows 11 (NT build 26200), x86-64
Python	3.11.9 (MSC 19.38, 64-bit)
Java	OpenJDK 21.0.10 (Microsoft build), HotSpot 64-bit VM
C compiler	GCC 16.1.0 (MinGW-W64), flags `-O2 -std=c11`
Libraries	NumPy 2.2.6 (input generation and correctness reference only), psutil 7.1.0 (memory and CPU sampling)
Input seed	42, values uniform in $[-1, 1)$, `float64` row-major

The machine is a shared laptop: no CPU affinity or idle-system guarantee was imposed. System CPU utilisation was sampled every 50 ms during the campaign (mean 24 %, maximum 100 %) and is reported as a limitation in Section 5.

3.4 Measurement protocol and budgets

- **Timing boundary:** multiplication kernel only. Input loading, allocation of C, checksum computation and output happen outside the timed region. Monotonic clocks: `time.perf_counter_ns` (Python), `System.nanoTime()` (Java), `QueryPerformance Counter` (C).
- **Warm-ups:** excluded from the statistics; counts above. One warm-up also precedes Python and C measurements for cache symmetry.
- **Repetitions:** 5 measured samples per configuration; very short kernels are batched so that each timing spans at least 0.2 s, and the per-multiplication time is the batch time divided by the batch size.
- **Estimator:** median with interquartile range (IQR); all raw samples are kept in `data/results/raw_timings.csv`.

- **Budgets:** 60 s per multiplication and 4 GiB peak resident memory, declared before the run. Every configuration first runs a feasibility probe (one warm-up kernel, then one timed kernel; one retry if the probe completed above budget). Configurations whose probe exceeded the budget, or whose probe was killed at the 150 s deadline before finishing the first kernel, are reported as timeouts *without* an invented execution time.
- **Memory metric:** peak resident set size of the child process, sampled every 50 ms by the orchestrator with psutil; Java additionally reports its managed heap after the run. These are different metrics and are not compared as if they were identical.

4 Results

4.1 Runtime

Table 2 reports the median and IQR percentage of the five measured samples for every configuration; Figure 1 shows the same data on log–log axes with the cubic reference. **TO** marks the declared timeout.

Table 2: Kernel time per multiplication: median in ms with IQR in % of the median (5 repetitions, warm-ups excluded). TO = exceeded the 60 s budget (no time invented).

n	Python	Java	C
10	0.167 (0.8)	0.00163 (2.7)	0.00127 (2.1)
50	17.14 (2.1)	0.168 (1.2)	0.102 (0.6)
100	135.9 (0.7)	1.51 (2.3)	0.971 (1.2)
200	1195 (1.6)	12.17 (0.1)	9.95 (0.8)
500	24 193 (10.2)	610.5 (36.5)	213.9 (5.1)
1000	TO	5246 (1.0)	2270 (3.9)
2000	TO	TO	TO

Fitting $\log T = a + p \log n$ by ordinary least squares over the sizes $n \geq 100$ (smaller sizes are dominated by fixed call overheads) gives the exponents in Table 3; all fits are close to $p = 3$ but systematically above it, with $R^2 \geq 0.995$.

Table 3: Fitted growth exponents over $n \geq 100$ (hypothesis H1 predicts $p = 3$).

Language	exponent p	R^2	sizes used
Python	3.223	0.9998	100, 200, 500
Java	3.640	0.9952	100, 200, 500, 1000
C	3.366	1.0000	100, 200, 500, 1000

The constant factors (hypothesis H2) are summarised in Figure 2: Python is 113–169× slower than C and 40–102× slower than Java over the shared feasible range; Java is 1.2–2.9× slower than C. Peak throughput reaches 2.46 GFLOP/s (C, $n = 50$), 1.49 GFLOP/s (Java, $n = 50$) and 0.015 GFLOP/s (Python, $n = 50$), against the 8.6 GFLOP/s peak a 2.8 GHz core could theoretically sustain with one multiply–add per cycle; the decline of all three curves at large n reflects memory-bandwidth and cache-miss costs rather than the arithmetic.

4.2 Variability

Figure 3 shows the IQR as a percentage of the median and the full distribution of the five samples. Most configurations are stable (IQR $\leq 5\,\%$); the exception is Java at $n = 500$ (IQR

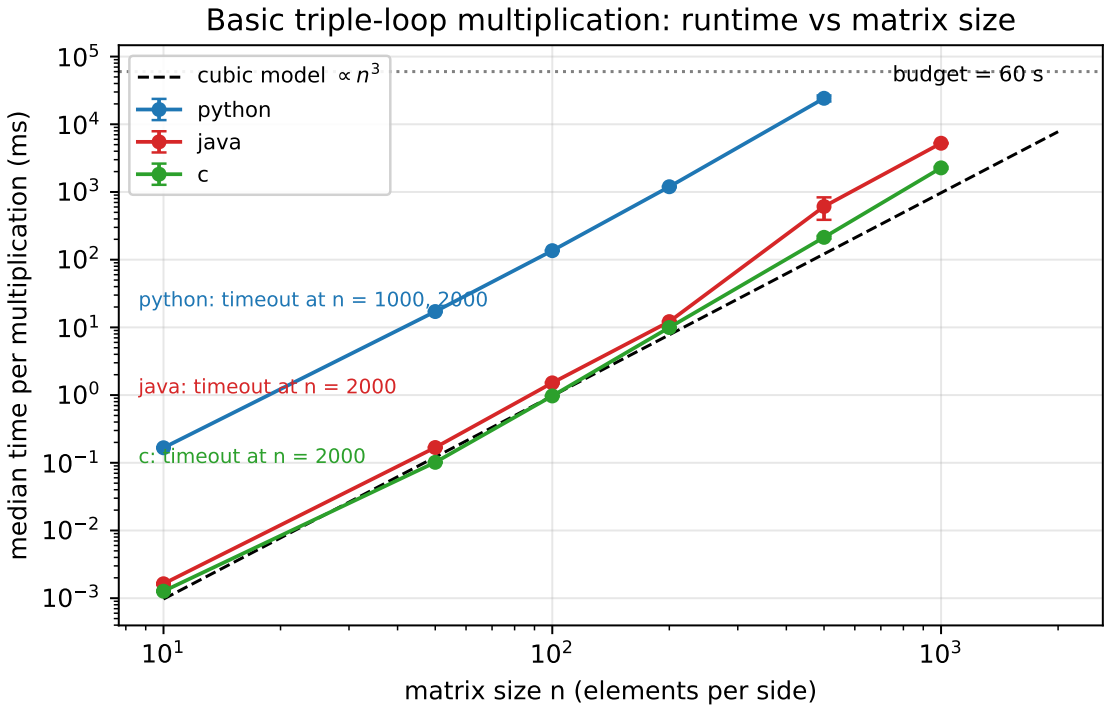


Figure 1: Median kernel time per multiplication versus matrix size (log–log, IQR error bars) against the n^3 reference line. The dotted line is the 60 s budget; timeout configurations are annotated but not plotted, because assigning them a runtime would fabricate data. Python follows a parallel line about two orders of magnitude above C; Java and C converge towards the cubic reference at small n and bend above it from $n = 500$.

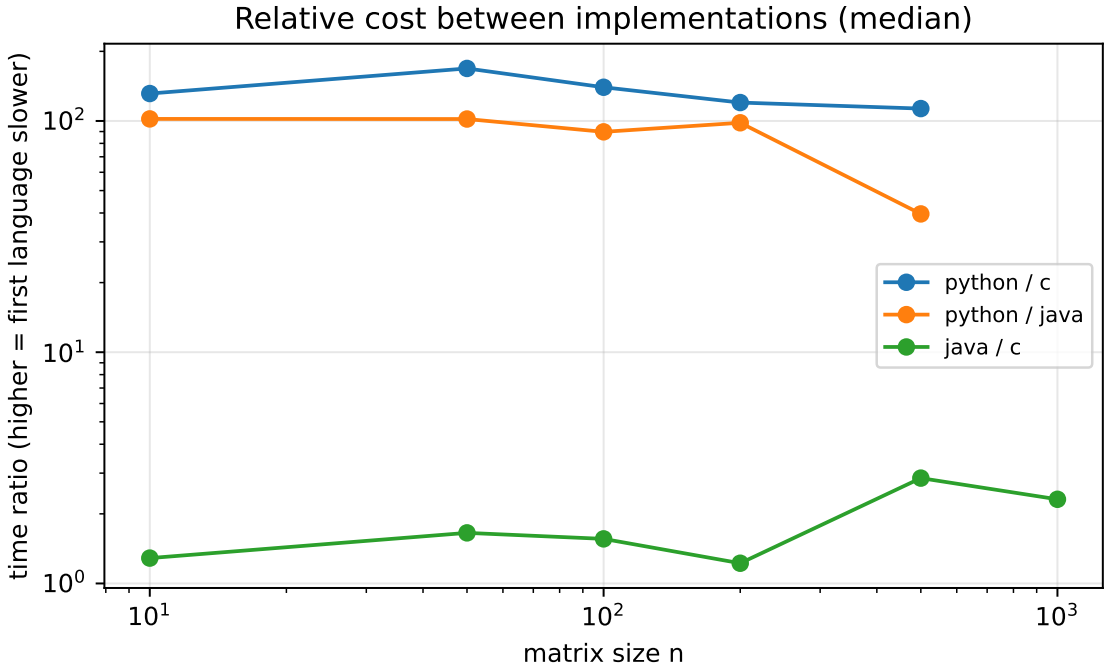


Figure 2: Ratio of median times between implementations at shared sizes (log–log). Ratios are roughly constant in the cache-resident regime; the Java/C ratio rises above $n = 200$ as Java's larger memory footprint and GC activity accumulate, while Python's gap to C slowly narrows (169× at $n = 50$ to 113× at $n = 500$) because fixed per-call overheads inflate the very small sizes.

36.5 %), which coincided with the highest background CPU activity of the campaign, and Python at $n = 500$ (10.2 %). No sample was discarded; the median is robust to these bursts.

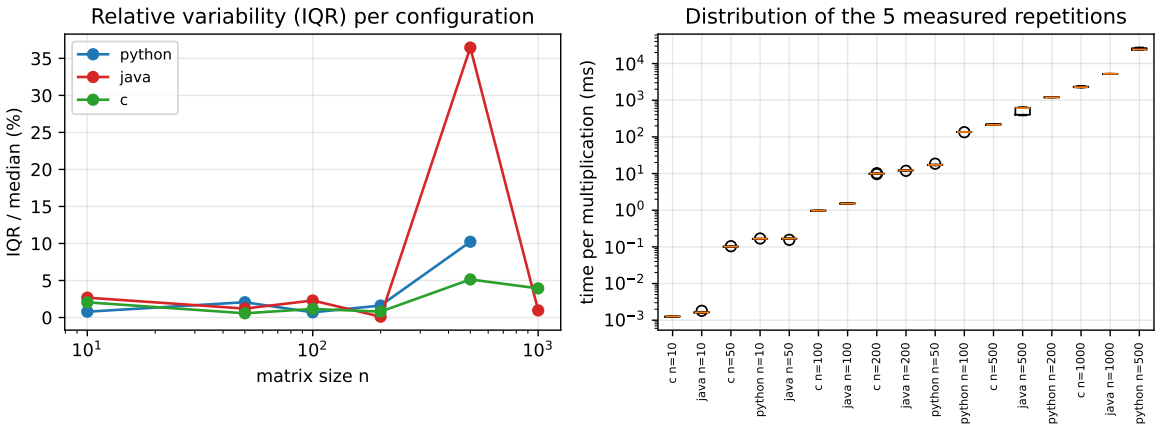


Figure 3: Left: IQR of the five repetitions relative to the median, per configuration. Right: box plot of the same samples (log scale). The spread measures machine noise (background desktop load, frequency scaling), not algorithmic variance, because each configuration is deterministic.

4.3 Memory

Table 4 reports peak resident memory (the named metric of this study). C tracks the $24n^2$-byte model plus a few megabytes of process overhead; Python adds about 16–20 MB of interpreter baseline; Java carries a ∼40 MB JVM baseline that grows to 106 MB at $n = 1000$ (heap plus JIT code, class metadata and threads). This supports H3: quadratic growth plus a constant per-language overhead. The theoretical raw storage is 6 MB at $n = 500$ and 24 MB at $n = 1000$.

Table 4: Peak process resident memory in MB (psutil, 50 ms sampling). "–" = configuration not run (budget exceeded).

n	Python	Java	C
10	18	74	4
50	16	42	4
100	20	43	4
200	21	46	5
500	46	54	10
1000	–	106	27
2000	–	–	–

4.4 Warm-up effect (Java)

Figure 5 replays the excluded warm-up kernels of the Java $n = 10$ run: the first multiplication of a fresh JVM takes 62.5 µs (mostly interpreted), while the steady state is 2.10 µs – a 30× difference inside a single process. Including warm-ups would have biased every Java number upwards; excluding them, as the protocol requires, does not hide compilation cost but reports the steady-state behaviour that the other two languages exhibit from their first call.

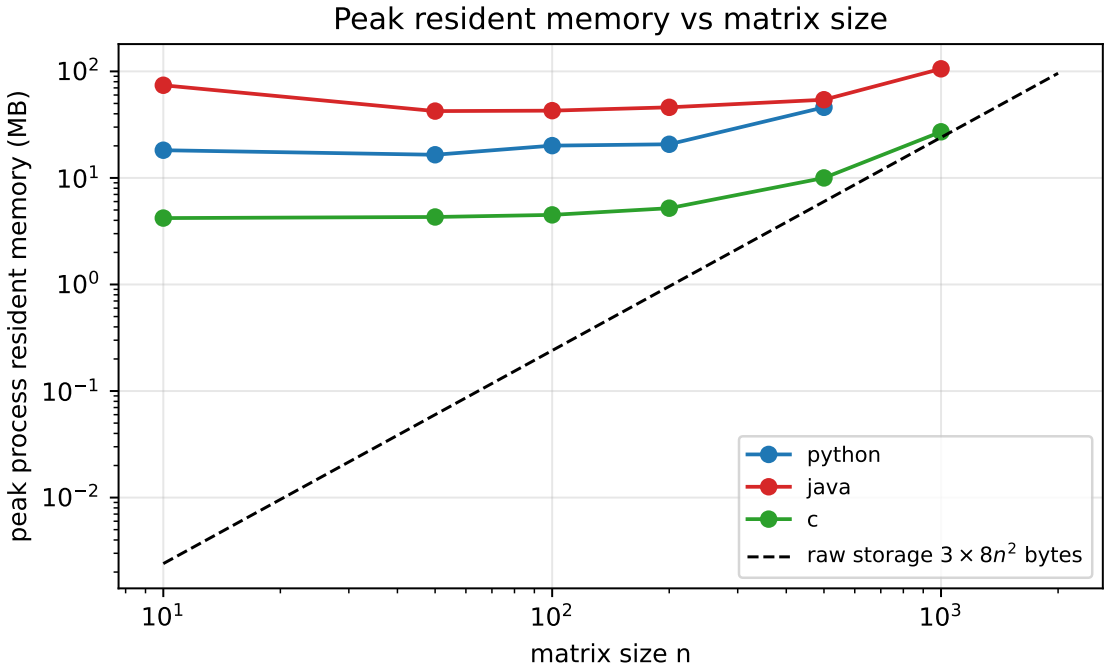


Figure 4: Peak resident memory versus n (log–log) against the raw-storage line $3 \times 8n^2$ bytes. The vertical offset of each curve is the runtime overhead of the language; the slopes match the quadratic model at large n.

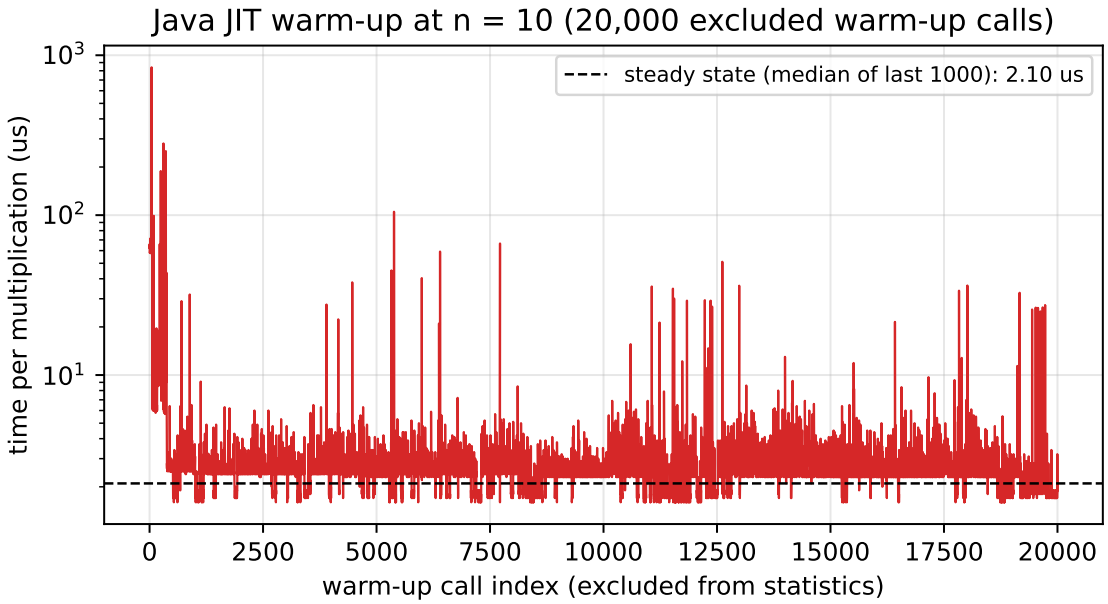


Figure 5: Java per-call time during the 20 000 excluded warm-up calls at $n = 10$ (log scale). Spikes are JIT recompilation and garbage-collection pauses; the dashed line marks the steady-state median of the last 1000 calls (2.10 μs).

Table 5: Feasibility limits (budget: 60 s per multiplication, 4 GiB memory).

Language	Largest feasible n (median)	Over budget	Killed before 1st kernel
Python	500 (24.2 s)	–	1000, 2000
Java	1000 (5.2 s)	–	2000
C	1000 (2.3 s)	2000 (probe: 77.0 s, 75.9 s)	–

4.5 Limits under the declared budgets

Python at $n = 1000$ and both Python and Java at $n = 2000$ were terminated by the 150 s probe deadline before the first kernel completed, so no runtime exists for them. C at $n = 2000$ completed its probe twice (77.0 s and 75.9 s, consistent to 1.4 %) and therefore is a measured *over-budget* result: the memory hierarchy at 96 MB of working matrices pushes the naive kernel past the budget even in native code. All four timeout outcomes are results in themselves: they locate the feasibility frontier required by the assignment.

5 Discussion

Answer to the research question. Within the tested scope, size affects runtime cubically with language-specific constants: over $n \geq 100$ the fitted exponents are 3.22 (Python), 3.64 (Java) and 3.37 (C), all close to the algorithmic n^3 model but above it because the working set outgrows the caches. Python pays a 113–169× constant relative to C (interpreted dispatch and boxed floats), Java only 1.2–2.9× after JIT compilation. Memory grows quadratically plus a constant runtime overhead (H3 confirmed: C $\approx 24n^2 + 4$ MB, Python + 16 MB, Java + 42 MB).

Why the exponents exceed three. Between $n = 500$ and $n = 1000$ C grows 10.6× instead of 8×: B grows from 2 MB (fits the last-level cache) to 8 MB and the B-column access stride doubles to 8 kB, increasing cache and TLB misses – a memory-system effect, not an algorithmic one. Java's larger exponent (3.64) includes GC and heap effects and the noisy $n = 500$ point (IQR 36.5 %).

Why the curves are not perfectly parallel. Small sizes are inflated by fixed costs that differ per language (function-call and interpreter overheads are large relative to $n = 10$ work), so ratios drift: Python/C falls from 131× at $n = 10$ to 113× at $n = 500$ while Java/C rises from 1.29 to 2.85 as Java's footprint grows.

Limitations.

1. **Shared, uncontrolled machine:** background load averaged 24 % (max 100 %) during the campaign; the largest observed IQR is 36.5 %. Medians with five samples mitigate but cannot eliminate this noise. The campaign could be repeated on an idle or pinned system.

2. **Single machine, single input set, single seed:** results do not generalise to other hardware, value distributions or $m \neq n$ shapes.

3. **Compiler settings:** GCC `-O2` may auto-vectorise the C kernel with the default x86-64 SSE2 target; this is fixed and documented, and it is one reason the C constant is small. It does not change the algorithmic complexity.

4. **Metric heterogeneity:** peak resident memory, managed heap and raw array storage are distinct quantities and are reported as such.

5. **Timeouts:** for killed probes no runtime exists; none was interpolated from smaller sizes.

Following the assignment guidance, no claim is made that one language is universally faster; the comparison is scoped to these implementations, flags, inputs and hardware [3, 4].

6 Conclusions

Within the tested scope the study establishes that: (1) the basic $O(n^3)$ triple-loop multiplication grows close to cubically in all three languages ($p = 3.22$–3.64 for $n \geq 100$), with super-cubic deviations appearing once the matrices leave the last-level cache; (2) language constants dominate the absolute times – Python 113–169× C, Java 1.2–2.9× C – while Java's JIT warm-up is a real, measurable effect that must be excluded from steady state; (3) peak resident memory follows $24n^2$ bytes plus a constant runtime overhead; and (4) under a 60 s budget the largest feasible sizes are $n = 500$ (Python) and $n = 1000$ (Java, C), with $n = 2000$ exceeding the budget in all three languages – in C a measured 76 s, in Python and Java a deadline kill without a runtime.

What remains unresolved: behaviour on an idle or pinned system, other value distributions and rectangular shapes, and the optimisation questions (blocking, loop reordering, sparse storage) that Assignment 2 addresses.

7 Reproducibility materials

Repository: `individual-assignments-bigdata` (GitHub; state the exact commit when submitting). Layout: `scripts/` (input generation, correctness validation, build, environment snapshot, benchmark orchestrator), `src/{python,java,c}/` (implementations), `data/inputs/` (generated matrices + SHA-256 manifest), `data/results/` (`raw_timings.csv`, `configs.csv`, `correctness_report.csv`, `summary.csv`, `limits.csv`, `environment.json`, per-process JSON logs), `notebooks/01_analysis.ipynb` (executed; regenerates every table and figure) and `reports/` (this PDF and `figures/`).

Exact commands:

```
python scripts/generate_inputs.py
python scripts/generate_inputs.py --check
python scripts/check_correctness.py
python scripts/collect_env.py
python scripts/run_benchmark.py
jupyter nbconvert --to notebook --execute notebooks/01_analysis.ipynb
cd reports && latexmk -pdf Assignment01_DelgadoValera_Report.tex
```

Generative-AI usage statement

Following the course policy, the author discloses that the AI assistant **opencode** (model `opencode/mimo-v2.6-flash-free`) was used, under the author's direction, for planning the measurement harness, drafting and debugging parts of the scripts, generating the analysis notebook and figures, and revising the English of this report. The author reviewed all generated code, executed and inspected the correctness tests, the benchmark campaign and every figure, verified the numbers quoted here against the raw CSV files, and is responsible for the measurements, the conclusions and any errors.

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